

CRIMEFIR

FIR Intelligence & Crime Pattern Detector

Reads raw FIR text, auto-drafts the NCRB crime classification, links FIRs across police stations and districts by hard evidence, flags repeat-offender groups and writes station crime briefs.

Team Code & Chaos · Track 4: AI & Predictive · Problem Statement 10 · IBM x NFSU Bob Hackathon

IBM Granite on watsonx.ai · IBM Granite Embedding · Laya · NetworkX · MCP server for IBM Bob

Crores of FIRs, no intelligence layer

- UP Police CCTNS alone holds 3+ crore digitised FIRs, with no NLP layer. Pattern analysis is manual.
- Jamtara-style gangs reuse the same phones and mule accounts against victims in many districts.
- Each victim files at their own station; every FIR writes 98765 43210 / +91-9876543210 / 09876543210 differently.
- Crime type and method (NCRB I.I.F.-II) are filled in by hand, later; the FIR itself has no crime-type field.
- Result: inter-district links are never surfaced; money in mule accounts is gone before anyone connects cases.

Investigating officer

cannot see their case is the 6th of a ring

Station House Officer

no timely picture of rising crime / active gangs

Victims

₹2,140 cr lost to digital-arrest scams alone (2024)

Evidence first, AI where it helps, a human decides

1 Rules: hard evidence

phones (all Indian formats), bank accounts, UPI IDs, IMEIs, vehicles, handles, with role: offender / property / complainant

2 Laya: System 1 (local GPU)

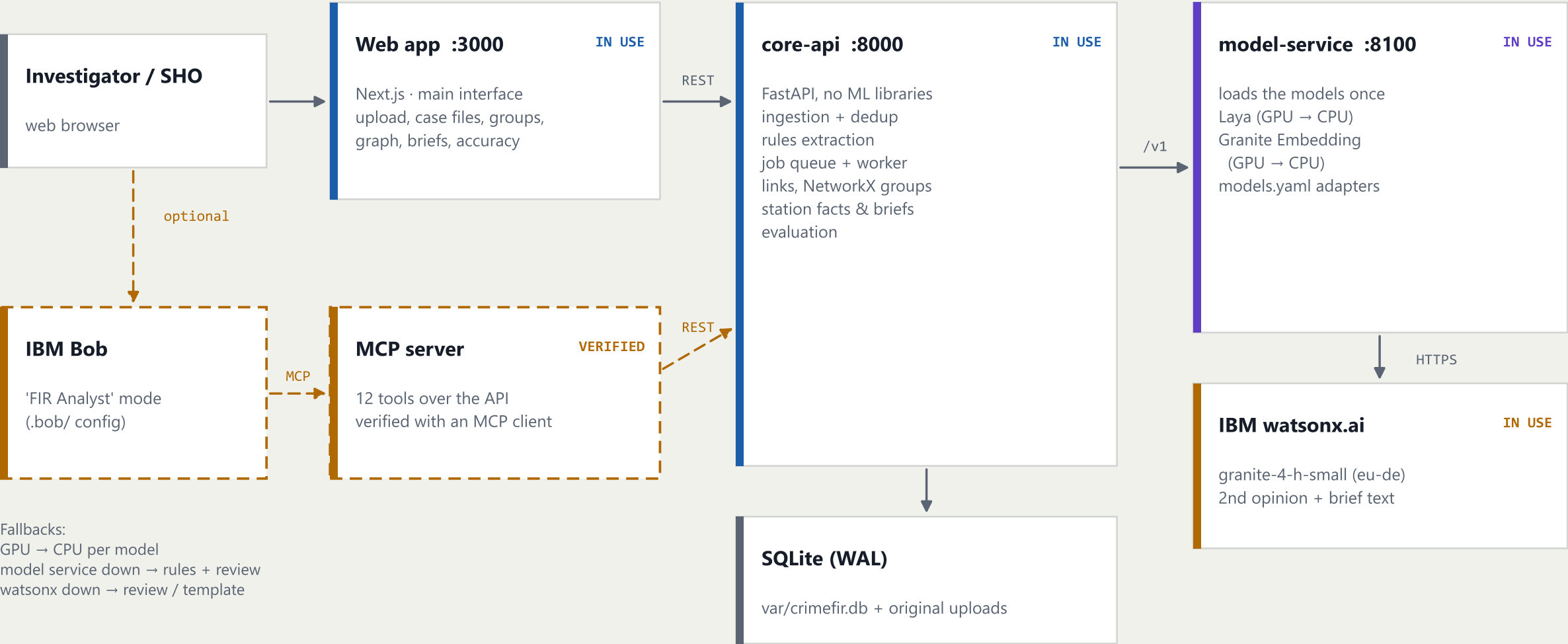
crime type (17) → category (4), 24 method flags, victim gender, with calibrated confidence; accepted if confidence ≥ 0.40

3 IBM Granite: System 2

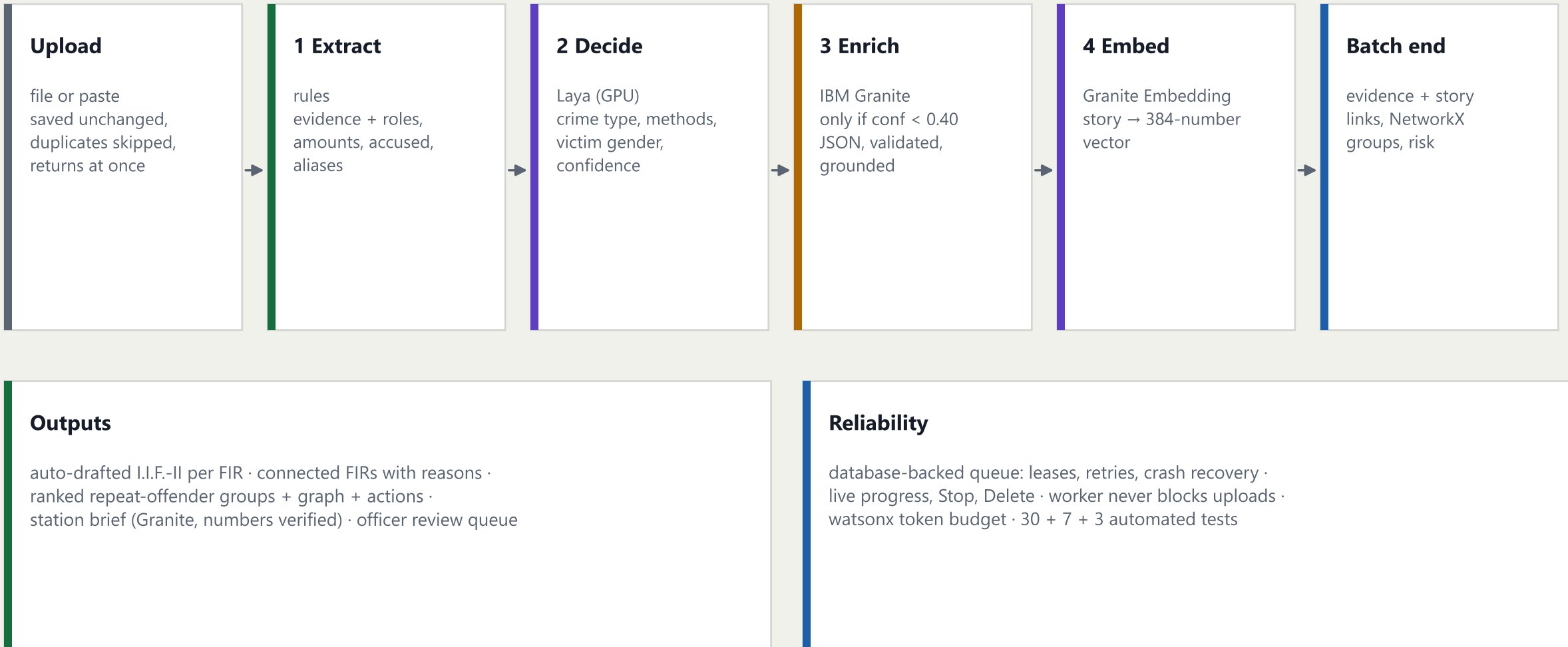
granite-4-h-small on watsonx.ai, only for ~13% unsure FIRs; strict JSON, validated, names must appear in the FIR

PS10 ASKS	CRIMEFIR DELIVERS
Categorise each FIR by crime type	Auto-drafted NCRB I.I.F.-II: 4 categories, 17 crime types
Extract accused, location, MO, victim profile	Accused + aliases, place, 24 method flags, victim, loss
Detect repeat-offender signatures	Evidence links + NetworkX groups with a transparent risk score
Station trend summary + flagged offender list	Number-verified Granite brief + ranked repeat-offender groups

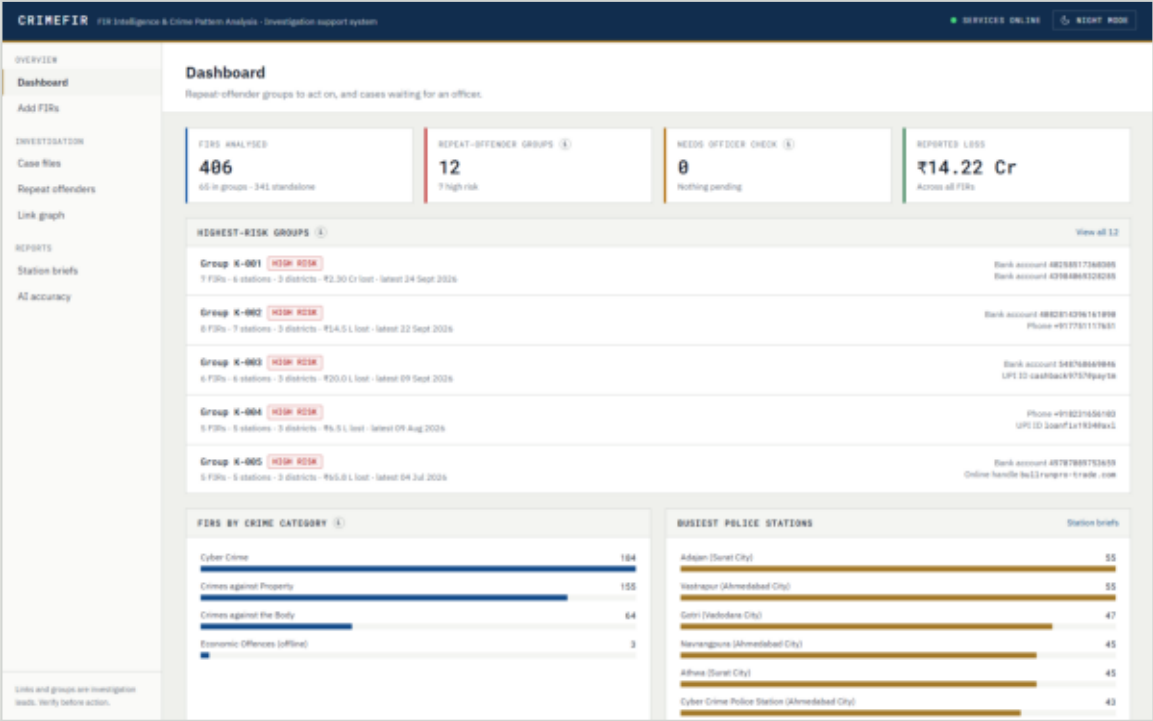
Solid = in use and verified. Dashed = implemented; the MCP server is verified end-to-end with an MCP client (smoke_test.py), but a live IBM Bob session has not been done yet.



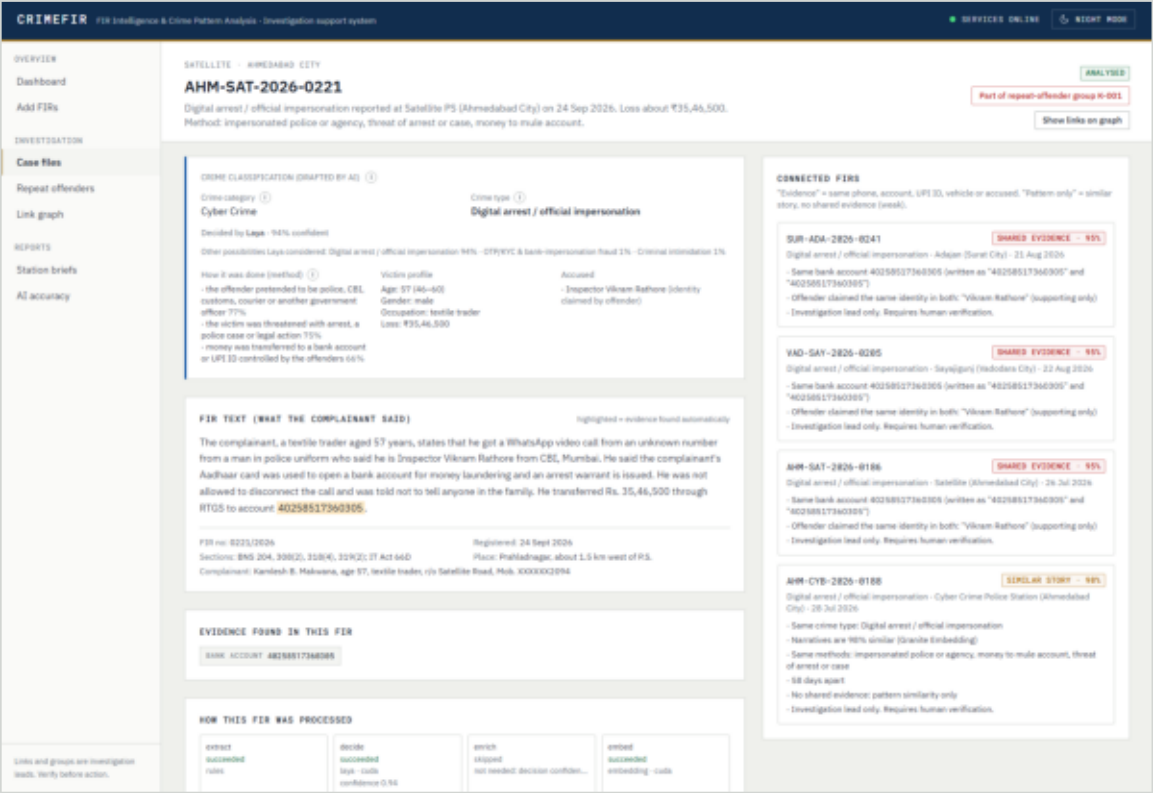
What happens to one FIR



What an investigator sees

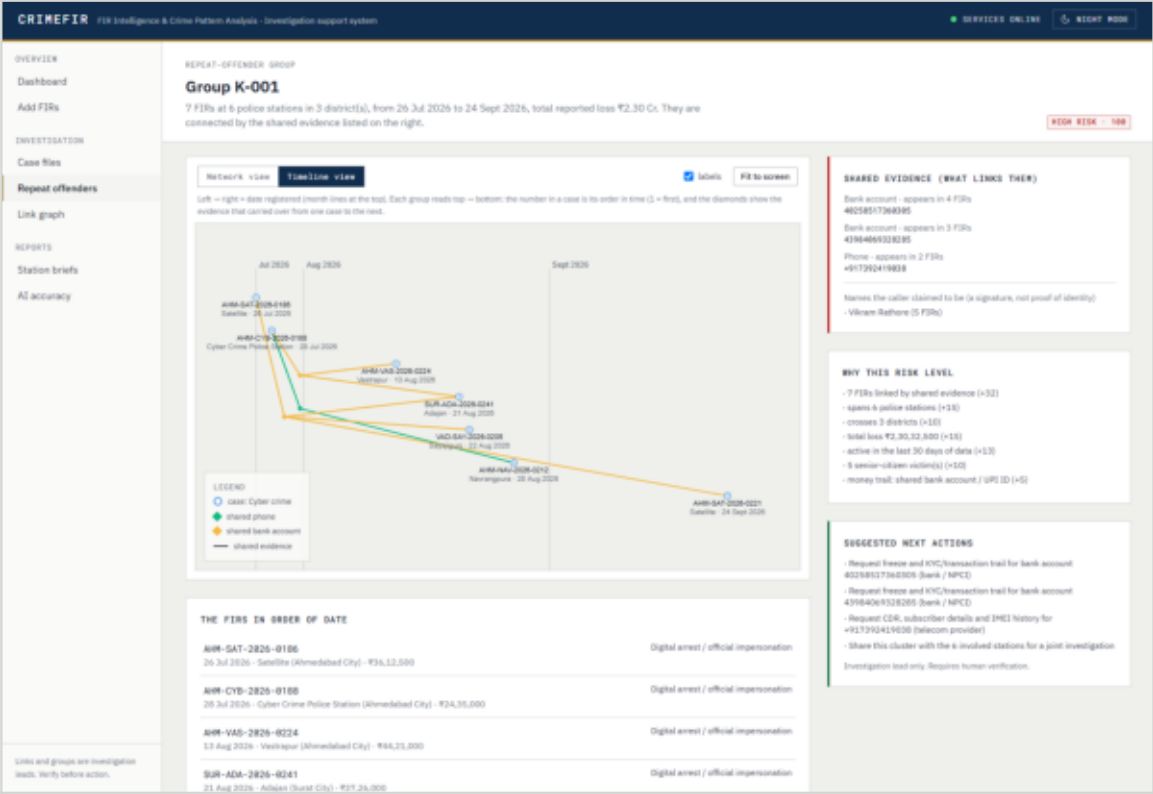


Dashboard: groups to act on first

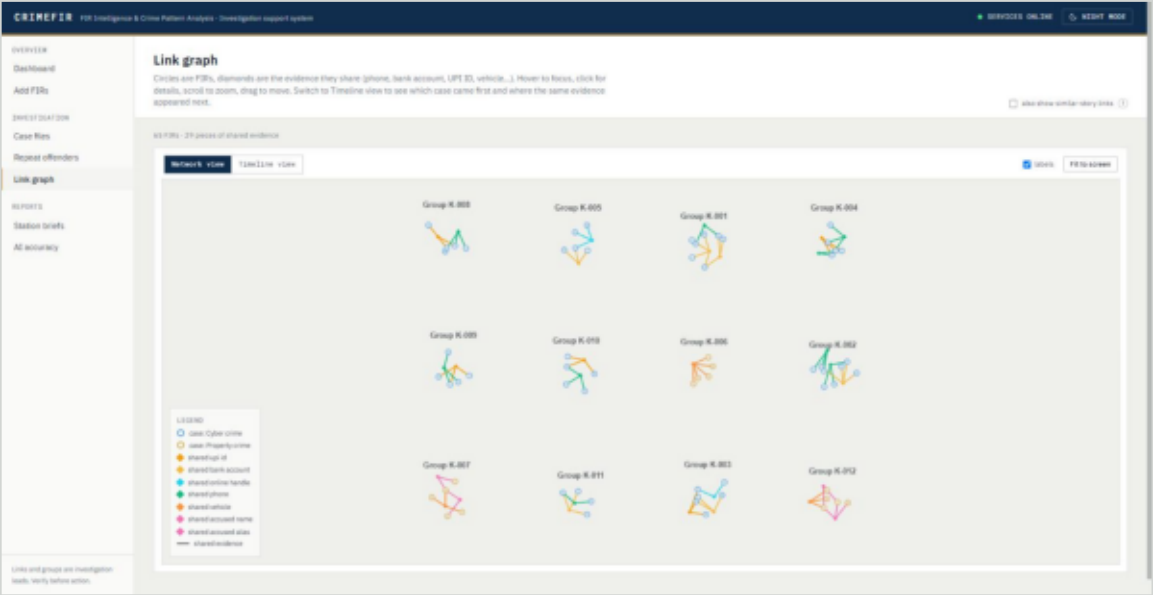


Case file: auto-drafted I.I.F.-II, evidence, connected FIRs

Repeat offenders across stations



One group: first case → later cases sharing the evidence



All 12 groups, coloured by shared evidence

Station briefs and measured accuracy

OVERVIEW

Dashboard

Add FIRs

INVESTIGATION

Case files

Repeat offenders

Link graph

REPORTS

Station briefs

AI accuracy

CRIMEFIR

FIR Intelligence & Crime Pattern Analysis - Investigation support system

SERVICES ONLINE

NIGHT MODE

Station briefs

Pick a police station and a period (default: the last 30 days of data). You see what happened there compared with the period before, and can generate a short written brief for the Station House Officer.

Cyber Crime Police Station (Ahmedabad City)mm/dd/yyyymm/dd/yyyy

Showing Cyber Crime Police Station (Ahmedabad City) from 28 Aug 2026 to 26 Sept 2026, compared with the same number of days just before.

FIRs THIS PERIOD

8

+33% vs period before (6)

MONEY LOST

₹34.7 L

VICTIMS Aged 18+

0

REPEAT-OFFENDER GROUPS ACTIVE HERE

0

NEED OFFICER CHECK

0

NUMBER OF FIRs BY CRIME TYPE

this period (number in brackets - period before)

Online job / task fraud (3)

3

OTR/KYC & bank impersonation fraud (2)

2

Fake customer-care / helpline fraud (2)

2

Digital arrest / official impersonation (1)

1

Increasing: Online job / task fraud 1→3 - Fake customer-care / helpline fraud 0→2

HOW CRIMES WERE COMMITTED (MOST COMMON)

the victim was sent a link, APK or fake app or website (7)

money was transferred to a bank account or UPI ID controlled by the offenders (6)

the victim paid an advance, fee, charge or deposit before receiving something (3)

the victim was threatened with arrest, a police case or legal action (3)

the victim was offered a job or paid online tasks (3)

REPEAT-OFFENDER GROUPS WITH FIRs HERE

none

SAFE EVIDENCE ALSO SEEN AT OTHER STATIONS

none

WRITTEN BRIEF FOR THE STATION HOUSE OFFICER

Input the numbers above. Output a short text written by IBM Granite (watsonx.ai), accepted only if every number in it matches the data.

Generate brief

28 Aug 2026 - 27 Sept 2026 - written by IBM Granite (watsonx.ai) - all numbers checked against the data - created 27 Sept 2026

Ahmedabad City Cyber Crime Police Station reported 8 First Information Reports (FIRs) in the period from 2026-08-28 to 2026-09-26, a 33% increase from the previous period. The total loss amounted to ₹34,65,500. Notably, online job/task fraud and fake customer-care/helpline fraud are rising, with counts increasing from 1 to 3 and from 0 to 2, respectively. The primary methods used by offenders include sending victims links, APKs, or fake apps/websites, transferring money to controlled bank accounts or UPI IDs, and threatening arrest or legal action. Repeat-offender clusters have not been identified, but links to other stations are pending verification. Focus areas should include monitoring and investigating the rising fraud types and methods.

Links and groups are investigation leads. Verify before action.

Brief written by IBM Granite, every number checked

OVERVIEW

Dashboard

Add FIRs

INVESTIGATION

Case files

Repeat offenders

Link graph

REPORTS

Station briefs

AI accuracy

CRIMEFIR

FIR Intelligence & Crime Pattern Analysis - Investigation support system

SERVICES ONLINE

NIGHT MODE

AI accuracy

Which AI did the work, how accurate it is, and which models are running. Accuracy is measured against a hidden answer sheet that comes with the test data.

1. WHO DECIDED THE CRIME TYPE

Laya (local AI) - decided on its own because it was at least 40% sure355 FIRs (87%)

IBM Granite on watsonx.ai - second opinion because Laya was less than 40% sure55 FIRs (15%)

IBM WATSONX.AI USAGE (GRANITE.LLK)

watsonx.ai is called for exactly two things: a second opinion on unsure FIRs, and writing station briefs.

Use

Calls

Tokens

Second opinion on unsure FIRs

111

1,617,349

Second opinion: asked again after an invalid answer

33

20,000

Writing station briefs

4

3,778

Total

128

1,90,927 of 2,58,800 monthly budget

2. HOW ACCURATE IT IS

Measured on 305 test FIRs that were never used to tune the system - 27 Sept 2026. FIRs uploaded without an answer sheet (e.g. your own) are processed normally but cannot be scored. The test data hides 12 gangs (groups of FIRs by the same offenders) and look-alike decoy cases.

CRIME CATEGORY CORRECT

93%

(Broad category (cyber, property, body, economic) matches the answer sheet.

EXACT CRIME TYPE CORRECT

83%

The specific type (e.g. 'OTR/KYC bank fraud', 'Chain snatching') matches. 17 possible types.

EVIDENCE FOUND

100%

Phones, accounts, UPI IDs, DMIDs, vehicles and handles found in the text. 100% of what was found is real (no false evidence).

ACCUSED NAMES FOUND

88%

Named accused / aliases picked up from the FIR. 2 FIR(s) got a wrong name.

GROUPS ARE CORRECT

100%

When two FIRs are put in the same repeat-offender group, how often they truly belong together (no wrong links).

HIDDEN GANGS FOUND

11 of 12

Gangs found completely, with exactly the right FIRs.

LOOK-ALIKE CASES WRONGLY GROUPED

0

Decoys have the same story as a gang but no shared evidence. They must NOT be grouped.

METHOD (NO) QUALITY

0.641

Score from 0 to 5 for detecting how the crime was done. The weakest part of the system.

Crime type accuracy by who decided: Laya (local AI): 80% of 265 FIRs - IBM Granite on watsonx.ai: 100% of 40 FIRs

3. MODELS IN USE

Links and groups are investigation leads. Verify before action.

Who decided each FIR, watsonx usage, accuracy

Team Code & Chaos • IBM x NFSU Bob Hackathon • Track 4 • PS10

Measured, not claimed

OUTPUT	RESULT
Crime category / crime type	93.4% / 83.0% (macro-F1 0.816)
Laya (accepted at ≥ 0.40) / IBM Granite (escalated)	80.4% on 265 FIRs / 40 of 40 correct
Hard evidence extraction	100% precision, 100% recall, 100% correct roles
Repeat-offender groups	100% precision, 93.7% recall, 11/12 gangs, 0 decoys
Accused names	88% found
Method flags (weakest output)	F1 0.64; 6 unreliable flags switched off
Team's first prototype, same 350-FIR file	found 1 of 12 repeated phones; CrimeFIR 12/12
Speed, RTX 3050 laptop (4 GB)	400 FIRs in ~6–8 min; live batch of 6 in 14 s

IBM technology: where it is used

IBM Granite on watsonx.ai

IN USE

granite-4-h-small, Frankfurt (eu-de)

used for exactly two things:

1. second opinion when Laya is
 < 40% sure (~13% of FIRs)
2. station brief text, numbers verified

IBM Granite Embedding

IN USE

granite-embedding-30m-english

local GPU, CPU fallback

story vectors for weak

'similar story' leads (never used
to flag anyone)

IBM Bob via MCP

SERVER VERIFIED

MCP server: 12 tools (search FIRs,
connected FIRs, groups, station
trends & briefs, ingest)
.bob/ 'FIR Analyst' mode + rules
live Bob session: at the hackathon

- A full 400-FIR run used about 90k watsonx tokens; a monthly token budget stops calls before the quota runs out.
- The MCP server is proven the way Bob uses it: started over stdio, 12 tools listed, calls return live data.
- With Bob: "Which high-risk groups touched Navrangpura this month, and what first?" → answers with FIR / group ids.

Impact, limits and next steps

- Works on existing FIR text: no change to how FIRs are written; outputs map to the NCRB forms.
- Surfaces inter-district rings in minutes: faster freeze requests on mule accounts, joint investigations.
- Scale path: PostgreSQL + pgvector, incremental linking, model-service replicas, batched Laya on a GPU server.
- Next: run CrimeFIR inside IBM Bob, authentication and roles, Indian-language FIRs, fuzzy name matching.

Honest limitations

synthetic (real-case-grounded) data · names without a surname are not provably one person · method flags weakest · story links weak · no authentication yet · IBM Bob not yet run live · every link is a lead that needs verification